MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, ADVANCED
SCHOTTKY TTL, SHIFT REGISTERS, CASCADABLE, MONOLITHIC SILICON

This amendment forms a part of MIL-M-38510/336A, dated 9 August 1983, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title: Delete and substitute as printed above.

1.2.1, under Circuit: Add "cascadable" after register.

1.2.3, under Letter: Delete "X" and substitute "2".

* 1.2.3, Case outline, add: "X C2-A (20-terminal, .350" x .350"), square chip carrier package)".

* 1.3, Thermal resistance: Delete in its entirety (as substituted in amendment 2) and substitute the following:

"Thermal resistance, junction-to-case (θ_{JC}):
Cases E, F, 2, and X - - - - - - - - - - - - - - (See MIL-M-38510, appendix C)".

1.3, after Junction temperature (T_J): Add "2/".

* At bottom of page, delete footnotes 2/ and 3/ (as added in amendment 2) and substitute the following:

"2/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening condition in accordance with method 5004 of MIL-STD-883."

PAGE 2

1.4, after Supply voltage: Add "V_{CC}".

1.4: Add "Width of clock pulse low, $t_{PW(CPL)}$ - - - - 6.0 ns minimum".

1.4, Setup time data high to clock pulse $t_{SD(H)}$: Delete "5.0 ns" and substitute "6.0 ns".

1.4, Setup time select high to clock pulse, $t_{SS(H)}$: Delete "10.0 ns" and substitute "10.5 ns".

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*The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
9	9
10	10
13	13
14	14

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PAGE 3

TABLE I, add the following:

Parameter	Symbol	Conditions $-55°C \leq T_C \leq +125°C$	Device type	Limits		Unit
				Min	Max	
Maximum frequency	f_{MAX}	V_{CC} = 5.0 V; C_L = 50 pF ±10%; (see figure 5)	All	80	---	MHz

Footnote 1/, delete and substitute:

"1/ Not more than one output should be shorted at a time."

PAGE 4

3.2.3, delete and substitute:

"3.2.3 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in the specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request."

3.6, delete and substitute:

"3.6 Marking. Marking shall be in accordance with MIL-M-38510."

TABLE II, Interim electrical parameters: Delete "(pre burn-in)".

TABLE II, Group A test requirements, under class B devices: Add "8".

PAGE 5

* 4.2a.(1): Delete "D or E" and substitute "D, E, or F".

4.2c., delete and substitute:

"c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510."

PAGE 6

* FIGURE 1: Delete " Case X" and substitute "Cases 2 and X".

PAGE 12

* TABLE III: Delete "Case X" and substitute "Cases 2 and X".

PAGE 15

4.4.3c.(1): Delete "D or E" and substitute "D, E, or F".

4.4.3c.(3), delete and substitute: "(3) Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883."

6.1: Delete in its entirety.

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PAGE 16

TABLE IV: Add "X" to circuit C column.

The margins of this amendment are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodians:
Air Force - 17
Army - ER
Navy - EC

Review activities:
Air Force - 11, 19, 85, 99
Army - AR, MI
Navy - OS, SH, TD
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

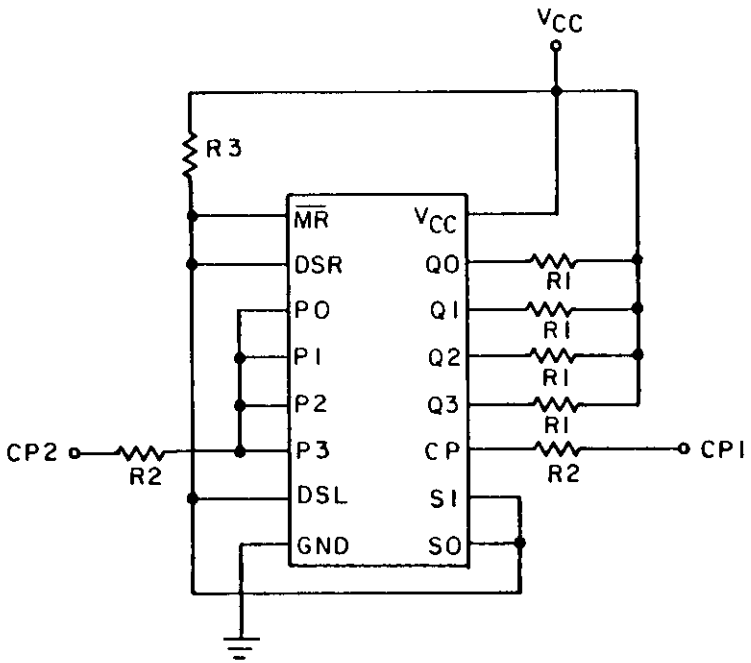
Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1040)

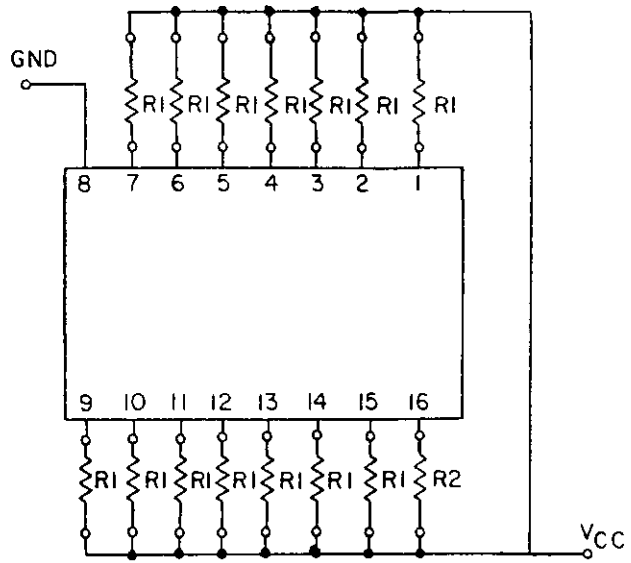
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Device type 01
Cases E, F, 2, and X
(Pins not referenced for cases 2 and X are N/C)



NOTES:
1. CP = 100 kHz ±50% square wave; duty cycle = 50 ±15%; CP2 = CP1 ÷ 2; V_{IL} = -0.5 V minimum to 0.8 V maximum: V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R1 = 240Ω ±5%; R2 = 51Ω ±5%; R3 = 1 kΩ ±5%.
3. V_{CC} and R4 shall be chosen to insure 5.5 V minimum is present at device V_{CC} terminal.

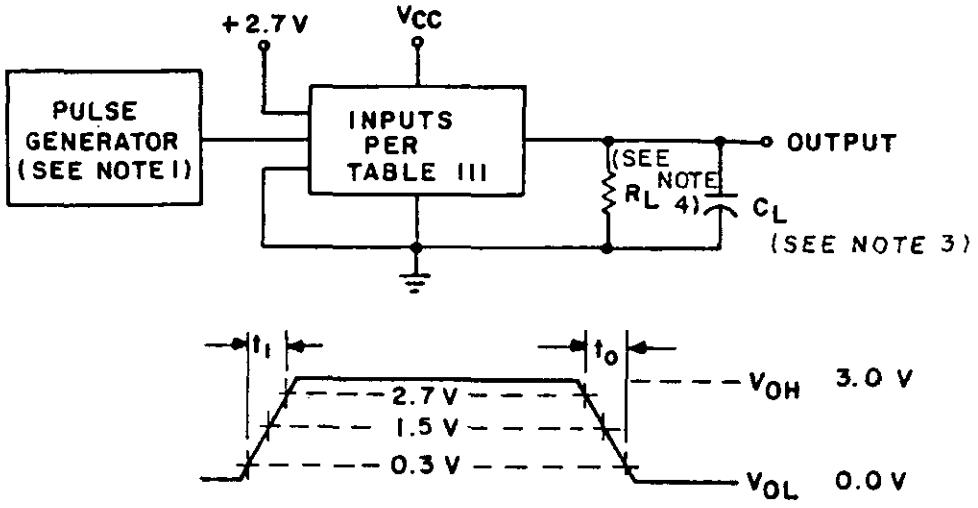
Device 01 excluding cases 2 and X
High-temperature (175°C) burn-in (optional circuit).



NOTES:
1. R1 = 2 kΩ ±5 percent, R2 = 2Ω
2. V_{CC} shall be high enough to ensure 5.0 V at the device terminal.

FIGURE 4. Burn-in and life test circuit.

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NOTES:
1. Pulse generator has the following characteristics: $t_1 = t_0 \leq 2.5$ ns, PRR $\leq$ 1 MHz, $Z_{OUT} \cong 50\Omega$.
2. Inputs not under test are at ground.
3. C_L = 50 pf ±10%, including scope probe, wiring and stray capacitance without package in test fixture.
4. RL = 499Ω ±5%.
5. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 5. Switching time waveform.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F / Cases 1/ 2,X	1 / 2	2 / 3	3 / 4	4 / 5	5 / 7	6 / 8	7 / 9	8 / 10	9 / 12	10 / 13	11 / 14	12 / 15	13 / 17	14 / 18	15 / 19	16 / 20	Measured terminal	Limits Min	Limits Max	Unit
			Test no.	MR	OSR	PO	P1	P2	P3	OSL	GND	SO	S2	CP	Q3	Q2	Q1	Q0	V_{CC}				
1 T_C = 25°C	I_{OS}	3011	53	5.5 V		5.5 V	0 V	0 V	0 V		GND	5.5 V	5.5 V	3/				0 V	5.5 V	Q0	-60	-150	mA
		"	54	"		0 V	5.5 V	0 V	"		"	"	"	"					"	Q1	"	"	"
		"	55	"		"	0 V	5.5 V	5.5 V		"	"	"	"	0 V	0 V	0 V		"	Q2	"	"	"
		"	56	"		"	0 V	0 V	"		"	"	"	"					"	Q3	"	"	"
	I_{DD}		57	0 V							"				2.5 V				4.5 V	Q0	60		"
			58	"							"					2.5 V			"	Q1	"		"
			59	"							"						2.5 V		"	Q2	"		"
			60	"							"								"	Q3	"		"
	I_{CC}		61	5.5 V	5.5 V	0 V	0 V	0 V	0 V	5.5 V	"	5.5 V	5.5 V	3/					5.5 V	V_{CC}		46	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C = 25°C	Functional tests 5/	3014	62	B	A	A	A	A	A	A	A	GND	A	A	A	L	L	L	L	6/	All outputs			
		"	63	A	A	B	B	B	B	B	B	"	"	"	A	H	H	H	H	"	"			
		"	64	"	B	A	B	B	B	B	B	"	"	"	B	L	L	L	L	"	"			
		"	65	"	A	A	A	A	A	A	A	"	"	"	A	H	H	H	H	"	"			
		"	66	"	B	B	B	B	B	B	B	"	"	"	B	L	L	L	L	"	"			
		"	67	"	A	A	A	A	A	A	A	"	"	"	A	H	H	H	H	"	"			
		"	68	"	B	B	B	B	B	B	B	"	"	"	B	L	L	L	L	"	"			
		"	69	"	A	A	A	A	A	A	A	"	"	"	A	H	H	H	H	"	"			
		"	70	"	B	B	B	B	B	B	B	"	"	"	B	L	L	L	L	"	"			
		"	71	"	A	A	A	A	A	A	A	"	"	"	A	H	H	H	H	"	"			
		"	72	"	B	B	B	B	B	B	B	"	"	"	B	L	L	L	L	"	"			
		"	73	"	A	A	A	A	A	A	A	"	"	"	A	H	H	H	H	"	"			
		"	74	"	B	B	B	B	B	B	B	"	"	"	B	L	L	L	L	"	"			
		"	75	"	A	A	A	A	A	A	A	"	"	"	A	H	H	H	H	"	"			
		"	76	"	B	B	B	B	B	B	B	"	"	"	B	L	L	L	L	"	"			
		"	77	"	A	A	A	A	A	A	A	"	"	"	A	H	H	H	H	"	"			
		"	78	"	B	B	B	B	B	B	B	"	"	"	B	L	L	L	L	"	"			
		"	79	"	A	A	A	A	A	A	A	"	"	"	A	H	H	H	H	"	"			
		"	80	"	B	B	B	B	B	B	B	"	"	"	B	L	L	L	L	"	"			
		"	81	"	A	A	A	A	A	A	A	"	"	"	A	H	H	H	H	"	"			
		"	82	"	B	B	B	B	B	B	B	"	"	"	B	L	L	L	L	"	"			
		"	83	"	A	A	A	A	A	A	A	"	"	"	A	H	H	H	H	"	"			
		"	84	"	B	B	B	B	B	B	B	"	"	"	B	L	L	L	L	"	"			
		"	85	"	A	A	A	A	A	A	A	"	"	"	A	H	H	H	H	"	"			
		"	86	"	B	B	B	B	B	B	B	"	"	"	B	L	L	L	L	"	"			
		"	87	"	A	A	A	A	A	A	A	"	"	"	A	H	H	H	H	"	"			
		"	88	"	B	B	B	B	B	B	B	"	"	"	B	L	L	L	L	"	"			
		"	89	"	A	A	A	A	A	A	A	"	"	"	A	H	H	H	H	"	"			
		"	90	"	B	B	B	B	B	B	B	"	"	"	B	L	L	L	L	"	"			
		"	91	"	A	A	A	A	A	A	A	"	"	"	A	H	H	H	H	"	"			
		"	92	"	B	B	B	B	B	B	B	"	"	"	B	L	L	L	L	"	"			
		"	93	"	A	A	A	A	A	A	A	"	"	"	A	H	H	H	H	"	"			
		"	94	"	B	B	B	B	B	B	B	"	"	"	B	L	L	L	L	"	"			
		"	95	"	A	A	A	A	A	A	A	"	"	"	A	H	H	H	H	"	"			
		"	96	"	B	B	B	B	B	B	B	"	"	"	B	L	L	L	L	"	"			
		"	97	"	A	A	A	A	A	A	A	"	"	"	A	H	H	H	H	"	"			
		"	98	"	B	B	B	B	B	B	B	"	"	"	B	L	L	L	L	"	"			
		"	99	A	A	A	A	A	A	A	A	"	"	"	A	H	H	H	H	"	"			
		"	100	"	A	B	B	B	B	B	B	"	"	"	B	L	L	L	L	"	"			
		"	101	"	B	A	A	A	A	A	A	"	"	"	A	H	H	H	H	"	"			
		"	102	"	A	B	B	B	B	B	B	"	"	"	B	L	L	L	L	"	"			
		"	103	"	B	A	A	A	A	A	A	"	"	"	A	H	H	H	H	"	"			
		"	104	"	A	B	B	B	B	B	B	"	"	"	B	L	L	L	L	"	"			
		"	105	"	B	A	A	A	A	A	A	"	"	"	A	H	H	H	H	"	"			
		"	106	"	A	B	B	B	B	B	B	"	"	"	B	L	L	L	L	"	"			
		"	107	"	B	A	A	A	A	A	A	"	"	"	A	H	H	H	H	"	"			
		"	108	"	A	B	B	B	B	B	B	"	"	"	B	L	L	L	L	"	"			
		"	109	"	B	A	A	A	A	A	A	"	"	"	A	H	H	H	H	"	"			
		"	110	"	A	B	B	B	B	B	B	"	"	"	B	L	L	L	L	"	"			

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F / Cases 1/ 2,X / Test no.	1 / 2 / $\overline{MR}$	2 / 3 / DSR	3 / 4 / P0	4 / 5 / P1	5 / 7 / P2	6 / 8 / P3	7 / 9 / DSL	8 / 10 / GND	9 / 12 / S0	10 / 13 / S2	11 / 14 / CP	12 / 15 / Q3	13 / 17 / Q2	14 / 18 / Q1	15 / 19 / Q0	16 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
8	Repeat subgroup 7 tests at T_C = +125°C and T_C = -55°C.																						
9 T_C = 25°C	t_{PLH1}	3003	111 112 113 114	2.7 V " " "		2.7 V	2.7 V	2.7 V	2.7 V		GND " " "	2.7 V " " "	2.7 V " " "	IN " " "	OUT	OUT	OUT	OUT	5.0 V " " "	CP to Q0 CP to Q1 CP to Q2 CP to Q3	3.5 " " "	7.0 " " "	ns " " "
	t_{PHL1}	3003	115 116 117 118	" " " "		0 V	0 V	0 V	0 V		" " " "	" " " "	" " " "	" " " "	OUT	OUT	OUT	OUT	" " " "	CP to Q0 CP to Q1 CP to Q2 CP to Q3	" " " "	" " " "	" " " "
	t_{PHL2}	3003	119 120 121 122	IN " " "		2.7 V	2.7 V	2.7 V	2.7 V		" " " "	" " " "	" " " "		OUT	OUT	OUT	OUT	" " " "	$\overline{MR}$ to Q0 $\overline{MR}$ to Q1 $\overline{MR}$ to Q2 $\overline{MR}$ to Q3	4.5 " " "	12.0 " " "	" " " "
	f_{MAX}		123 124 125 126 127 128	2.7 V " " " " "	IN	IN	IN	IN	IN	IN	" " " " " "	2.7 V 0.0 V 2.7 V " " "	0.0 V 2.7 V " " " "	IN " " " " "	OUT OUT	OUT	OUT	OUT OUT	" " " " " "	Q0 Q3 Q0 Q1 Q2 Q3	100 " " " " "		MHz " " " " "
10	t_{PLH1}	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C and limits as shown.																			3.0	8.5	ns
	t_{PHL1}																				3.0	8.5	ns
	t_{PHL2}																				4.5	14.5	ns
	f_{MAX}																				80		MHz
11		Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																					

1/ For cases 2 and X pins not referenced are N/C.

2/ Apply 3.0 V min/5.5 V max, 0 V, 3.0 V min/5.5 V max to $\overline{MR}$ then 3.0 V min/5.5 V max, 0 V, 3.0 V min/5.5 V max to CP prior to making measurement.

3/ Apply 3.0 V min/5.5 V max, 0 V, 3.0 V min/5.5 V max to CP prior to making measurement.

4/ I_{IL} limits shall be as follows:

Parameter	Test no.	Limits in mA for circuits A	B	C
I_{IL1}	43 through 52	-.25/-.60	-.03/-.60	-.03/-.60

5/ A = 2.5 V minimum, B = 0.5 V maximum, X = don't care
H ≥ 1.5 V, L ≤ 1.5 V

6/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V

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